

TRANSLATING BETWEEN REPRESENTATIONS OF PROPORTIONAL RELATIONSHIPS



LESSON INTRODUCTION

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

LEARNING TARGETS

- I can determine the constant of proportionality of a proportional relationship.
- I can translate a given representation of a proportional relationship to a written description, table, or equation.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.3 Given a table, graph, or written description of a linear relationship, write an equation in slope-intercept form.

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

ESSENTIAL QUESTIONS

- How can you translate between the different representations of proportional relationships?
- How do you interpret the constant of proportionality in context?

LESSON NARRATIVE

In this lesson, students will translate between different representations of proportional relationships. In the 11.7.2 Guided Instruction, students use a given graph to complete a table of values, write an equation, and complete a written description of the proportional relationship. Students are then guided through practice where they analyze tables representing proportional relationships to write equations. Finally, students collaborate to match equations to their graphs before making connections to real-world contexts.

COMPONENT SUMMARY

11.7.1 WARM-UP (~5 MIN.)

Generate equivalent ratios

11.7.2 GUIDED INSTRUCTION (10-15 MIN.)

Translate between each representation of a proportional relationship

11.7.3 GUIDED PRACTICE (10 MIN.)

Practice translating between representations of proportional relationships

11.7.4 COLLABORATION (10-15 MIN.)

Match equations to equivalent graphs of proportional relationships

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Equation
- Constant of proportionality
- Unit rate
- Ratio
- Proportional
- Ordered pair

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse which order to divide when finding the constant of proportionality.

THINGS TO CONSIDER

- Consider opportunities to spiral concepts learned earlier in the unit in each of the components, including explaining how they know each relationship is proportional.
- Encourage students to utilize the vocabulary they have been learning throughout the unit when discussing contexts with their partners.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 11.7.2 Guided Instruction, students are prompted to compare two written descriptions. Encourage students to notice and wonder while comparing the two descriptions. How are they the same? How are they different? What do you wonder?

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.2.1 Multiple Representations**

Throughout this lesson, students are representing proportional relationships in a variety of ways. Specifically, in 11.7.2 Guided Instruction, where students translate a graph of proportional values to a table, equation, and written description. Translating to multiple representations allows students to develop a deeper understanding.

DIFFERENTIATE INSTRUCTION

Notice 11.7.2 Guided Instruction provides multiple entry points to engage learners. Organization of the values in tables and displays on the graph provide visual entry points, while partner discussion encourages students to verbally reason about the content before engaging in written explanations.

11.7.1 WARM-UP: EQUIVALENT RATIOS**DIFFERENTIATE INSTRUCTION**

Encourage students who finish early to create more ratios using numbers 1-20. Or change up the original ratio to offer more of a challenge.

**11.7.1 Warm-Up: Equivalent Ratios**

- Using the digits 1 to 9, fill in the boxes to make at least two ratios that are equivalent to $20 : 30$.

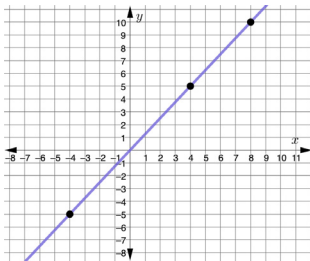
Answers will vary.

2	:	3
4	:	6



11.7.2 Guided Instruction: Translating Between Different Representations of Proportional Relationships

The graph of a proportional relationship is shown.



1. Complete the representations of the proportional relationship by filling in the blanks in the chart.

Table of Values		Equation	Written Description											
<table><tr><td>x</td><td>y</td></tr><tr><td>-4</td><td>-5</td></tr><tr><td>0</td><td>0</td></tr><tr><td>4</td><td>5</td></tr><tr><td>8</td><td>10</td></tr><tr><td>32.60</td><td>40.75</td></tr></table>	x	y	-4	-5	0	0	4	5	8	10	32.60	40.75	$y = \frac{5}{4}x$	As the value of x increases by <u>4</u> , the value of y increases by <u>5</u> .
x	y													
-4	-5													
0	0													
4	5													
8	10													
32.60	40.75													

2. Suppose in the previous relationship, x represents the price, in dollars, a store owner pays for merchandise from their manufacturer and y represents the price, in dollars, for which the store owner sells the merchandise.
- a. What is the constant of proportionality, $\frac{y}{x}$?

$\frac{y}{x} = \frac{5}{4}$ or 1.25

11.7.2 GUIDED INSTRUCTION: TRANSLATING BETWEEN DIFFERENT REPRESENTATIONS OF PROPORTIONAL RELATIONSHIPS

DIFFERENTIATE INSTRUCTION

Encourage students to see and use the graph as a visual representation when filling in the table of values and determining the equation. Consider having students mark the points on the graph that correspond to those indicated in the table of values to provide an additional visual connection between the representations.

Consider modeling the written description by showing that every time the x -value increases by 4, the y -value increases by 5 units on the graph.

11.7.2 GUIDED INSTRUCTION: TRANSLATING BETWEEN DIFFERENT REPRESENTATIONS OF PROPORTIONAL RELATIONSHIPS (CONTINUED)

TEACHER MOVES

Students may struggle to explain what the constant of proportionality means in context.

Remind students of the percentage relationships explored in Unit 7.

- Does this situation represent a discount or a markup?
- What is the percentage discounted or marked up?
- How can the constant of proportionality be interpreted in this context?

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

DIFFERENTIATE INSTRUCTION

Encourage students to see the relationship between each row in the table by drawing arrows to model the multiplication. By having students show that each x value is multiplied by the constant of proportionality, 1.25, to find the corresponding y -value, they can solidify understanding and find the equivalent ratio. They can also relate this to the equation representing the relationship.

- b. Explain what the constant of proportionality means in this context.

The store owner charges customers \$1.25 for every \$1.00 he pays for the merchandise.

- c. Complete the written description.

As the value of x increases by 4 dollars, the value of y increases by 5 dollars.

- d. Discuss with your partner what is different about the written description given the context compared to the written description in Problem 1.

- e. Explain which parts of the graph of this relationship do not make sense in the context.

It does not make sense to have a negative value on the graph because he will not be pay a negative amount of money nor charge the customers a negative amount of money.

- f. Use the equation to determine the sale price of an item the store owner purchases for \$32.60.

The sale price would be \$40.75.

- g. Write the coordinate pair of the point on the graph of this relationship that represents the item described in Part F.

(32.60, 40.75)

- h. Add a row to the table of values to represent this point.

Shown in the table of values above.

- i. Show that the ratio $\frac{y}{x}$ for this point is equivalent to the constant of proportionality.

$$\frac{40.75}{32.6} = \frac{5}{4} \text{ or } 1.25$$



11.7.3 Guided Practice: Translating Between Different Representations of Proportional Relationships

1. Each table represents a proportional relationship.
 - a. Find the constant of proportionality and write an equation that represents the relationship.

Table of Values	s	2	3	5	10
	P	8	12	20	40
Constant of Proportionality	4				
Equation	$P = 4s$				

- b. Find the constant of proportionality and write an equation that represents the relationship.

Table of Values	d	2	3	5	10
	c	6.28	9.42	15.7	31.4
Constant of Proportionality	3.14				
Equation	$c = 3.14d$				

2. It took Petra 5 minutes to fill a cooler with 8 gallons of water from a faucet flowing at a steady rate. Let w be the number of gallons of water in the cooler after t minutes.
 - a. Circle the two equations that represent the relationship between w and t .

$w = 1.6t$ $w = 0.625t$ $t = 1.6w$ $t = 0.625w$

- b. What does 1.6 represent in the context of this situation?
1.6 is the unit rate or constant of proportionality. It represents that for every minute the faucet is flowing, 1.6 gallons of water flow into the cooler.
 - c. What does 0.625 represent in the context of this situation?
0.625 is the other unit rate or constant of proportionality in this situation. It represents that it takes 0.625 of a minute to fill the cooler with 1 gallon of water.

3. In Citrus County, FL, a 6% sales tax is added to the price of any item sold.
 - a. Complete the table of values to represent the relationship between an item's price, p , in dollars, and the total cost to a customer, t , in dollars, after sales tax is applied.

p (\$)	1.50	4	20
t (\$)	1.59	4.24	21.20

- b. Write an equation that could be used to find the total cost, t , in dollars, of any item sold in Citrus County, after taxes.
 $t = 1.06p$

11.7.3 GUIDED PRACTICE: TRANSLATING BETWEEN DIFFERENT REPRESENTATIONS OF PROPORTIONAL RELATIONSHIPS

DIFFERENTIATE INSTRUCTION

Consider encouraging students to create a table of values, if necessary, comparing gallons and minutes in this proportional relationship. This will provide a visual entry point for students to determine how the variables are related.

DIFFERENTIATE INSTRUCTION

Challenge early finishers by having them write an additional equation to represent the relationship between the price and total cost.

11.7.4 COLLABORATION: MATCHING EQUATIONS TO GRAPHS

DIFFERENTIATE INSTRUCTION

Prompt students to identify the scale of each graph when matching to the equations.

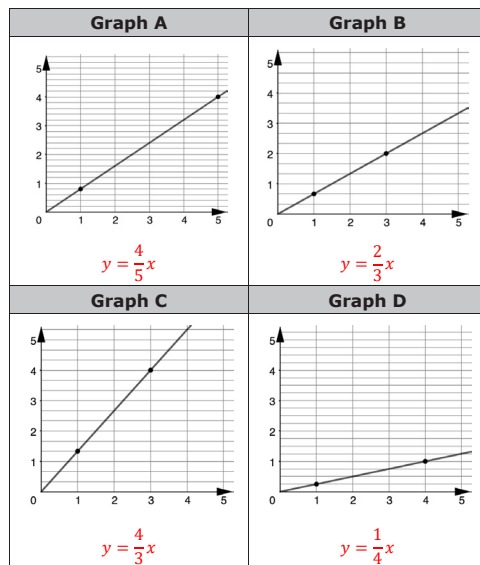


11.7.4 Collaboration: Matching Equations to Graphs

Several equations are shown.

$y = 2x$	$y = \frac{4}{5}x$	$y = \frac{1}{4}x$	$y = \frac{2}{3}x$	$y = \frac{4}{3}x$	$y = \frac{3}{2}x$
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1. Work with your partner to match each equation to the graph representing the same relationship. Write the equation in the space provided below each graph.



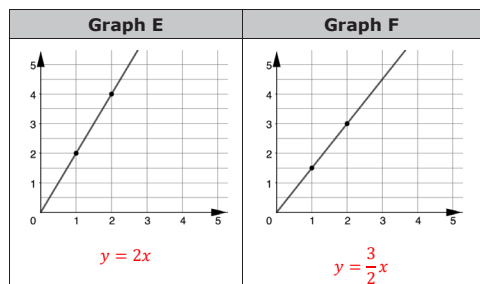
TEACHER MOVES

Highlight students' responses that mention:

- Finding constant of proportionality using $\frac{y}{x}$ and using that to find the equivalent equation.
- Finding the constant of proportionality, k , using the point $(1,k)$.

DIFFERENTIATE INSTRUCTION

Be prepared to talk students through converting percents to decimals, and then to fractions to completely answer question 4.



2. Choose one graph and explain how you determined the equation that matched the graph.
Answers will vary.
For each graph, determine the value of y when $x = 1$. This will be the constant of proportionality multiplied by x to result in y for each point on the line.
3. Share your explanation with your partner. If necessary, make edits to your explanation based on feedback from your partner.
4. A store is having a 20% off sale. With your partner, discuss which graph represents the relationship between the original price, in dollars, of an item, x , and its sale price, y , in dollars. Then, write a brief summary of your discussion.
Answers will vary.
A 20% off sale means that customers are paying 80% of the regular price. 80% as a fraction is $\frac{4}{5}$, which means that this situation was represented in Graph A, $y = \frac{4}{5}x$.

**11.7.5 Wrap-Up: Reflection**

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on translating proportional relationships to written descriptions, tables, or equations?

Answers will vary.

11.7.5 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.